

Effects of Motivation on Exploratory Behavior in College Students

Discussion Section 307: Samantha Baniel, Quincy Bianchi, Emma Cooley

Abstract *Contributed by: Samantha, Emma, Quincy*

A significant topic in decision-making is exploration vs exploitation. Deciding to discover new options or stick with old ones relates to the decisions people make every day. Research has shown that adults are more exploitative than children, especially in the presence of extrinsic motivation tactics such as timers, rankings, and badges. However, researchers have yet to examine the influence of a leaderboard on decision-making. Here, we examine how a leaderboard, an extrinsic motivator, influences exploratory or exploitative behavior in undergraduate college students. Using a between-groups design, each participant completed two rounds of the task. In condition one, participants did not know about the leaderboard until the beginning of the second round, whereas in condition two, they did know about it before the first round. In a 6x6 grid task, participants had 20 clicks per round to earn points through hidden rewards. We examined whether the presence of the leaderboard was associated with more exploitative behaviors. We also examined how performance in the first round influenced behavior in the second round, based on their distance from the highest possible score, as shown on the leaderboard. Our findings show that a leaderboard has the same effect on decision-making as other sensory stimuli.

Introduction *Contributed by: Samantha, Emma, Quincy*

A significant topic in decision-making is exploration vs exploitation ¹. Exploration in decision-making entails using a strategy with an uncertain payoff in the hope that it will be better than the current-best strategy, while exploitation refers to using known strategies to achieve a goal with minimal risk ². Given the prevalence of explore vs. exploit decisions in our daily lives (e.g., returning to your favorite restaurant or trying a new one), it is important to examine how different factors influence people's decision-making. This study uses existing research to explore the relationship between leaderboards and decision-making behavior.

There are many ways to experimentally study exploration vs. exploitation. A common method is to use a grid task, in which participants click squares on a spatial grid with randomly hidden rewards of varying values ^{3,4}. Using rewards to examine

decision-making behavior has been used in computer modeling reinforcement learning, demonstrating its robustness across modalities ⁵. Exploratory behaviors are represented by clicking in new spaces, whereas exploitative behaviors are represented by returning to the same square, knowing it reveals some level of reward.

The grid task has been adapted across research to address various questions, including how development coincides with changes in decision-making behavior. For example, to test children on the grid task, researchers used rewards that might actually entice children, such as fictitious creatures that gave them candy. This research showed that adults are more strategic and more likely to exploit, and less likely to explore, than children ^{3,6,7}. By this logic, college students, as adults, are more likely to engage in exploitative behavior without a specific reason to become more explorative. To test this idea in college students, the rewards hidden under the grid are applicable to them and to rewards they may encounter in their daily lives, such as a night out with friends or a good grade on an assignment.

Researchers have also largely examined the role of extrinsic motivation in decision-making behavior through visual factors such as timers, rankings, and badges, and their influence on exploitation ⁸. The more competitive a task seems, the more rewarding exploitative behavior becomes. Though these researchers briefly examined peer rankings, the literature as a whole has yet to examine the direct effect of a leaderboard on decision-making behavior.

The proposed research addresses the gap by testing college students on a grid task, examining their decision-making behaviors in the presence of a leaderboard. In this study, we aim to test whether the presence of a leaderboard increases exploitative behaviors on a game board in college students. By randomly assigning whether participants know the leaderboard in the first round, we aim to measure changes in individual behavior as participants begin without it and eventually gain it. In other words, whether participants' performance in the first round, as measured by the number of points earned, influences their behavior in the second round. Depending on the study's results, the findings can be applied to many real-life situations ⁹, allowing people to determine whether a leaderboard or other interactive extrinsic stimuli are beneficial for their decision-making and the goal at hand.

Building on prior literature, we predicted that participants who were cognizant of the leaderboard would exhibit more exploitative behavior, whereas those unaware of it would exhibit more exploratory behavior. A secondary hypothesis is that those who were closer to the total possible points earned in the first round would be more exploitative in the second round than those farther from it.

Methods *Contributed by: Samantha, Emma, Quincy*

Participants: The ideal sample size in the literature ranges from 100 to 200, which is sufficient for exploration-exploitation research with humans ^{4, 10}. We have recruited 67 undergraduate students at universities in the Midwest. Participants were randomly assigned to one of the two conditions, with 36 in condition 1 (unaware of leaderboard) and 31 in condition 2 (cognizant of leaderboard). The inclusion criteria was being an undergraduate student, excluding graduate students and all other members of the population.

Task: We created a game simulation in which all participants have an avatar who is a college student, and, through the task, they try to make choices that yield the most rewards for their avatar. The game board was a 6x6 grid, with each spot having a randomly chosen reward. Each round, participants had five minutes and 20 click opportunities. Participants were aware of the finite number of clicks for each round of the task. If participants did not use all their clicks, their session would time out after 5 minutes. Since they were limited to only 20 clicks per round, the participants needed to explore if they wished to find all rewards on the 36-square grid. Participants could click on any spot as many times as they wanted. Each reward had different point values to eliminate confounds from subjective value. Of the 36 squares on the grid, 18 were 5-point rewards, 12 were 10-point rewards, 4 were 15-point rewards, and 2 were 20-point rewards. These rewards were randomly assigned throughout the grid to avoid “hotspot” areas. The leaderboard was composed of our alpha and beta-tested scores. The leaderboard reset at the end of each round, so scores were not cumulative across rounds.

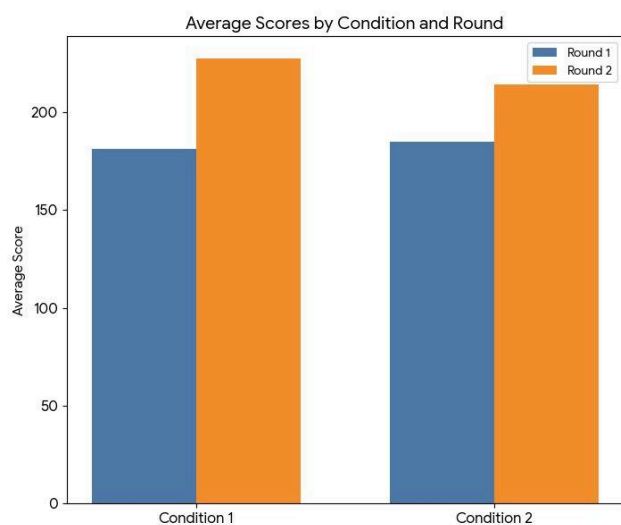
Variables: Extrinsic motivation is represented by the presence of a leaderboard. The independent variable is the presence of the leaderboard during the first round of the game. The dependent variables were their scores and the number of unique clicks.

Trial Structure: Participants completed two rounds of the game. Between the two rounds, they completed a distraction task: counting backward from 15 through typing. This distraction task was designed to ensure that participants did not memorize the grid's locations of rewards for the second round, since the grid did not change between rounds.

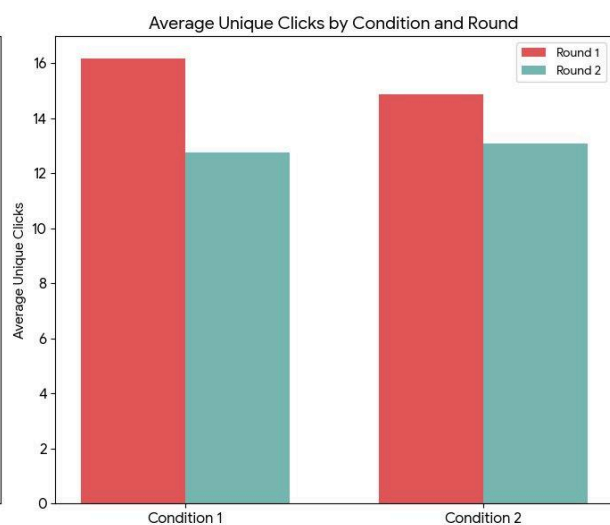
Between- or Within-Subjects Manipulations: This is a crossover design that compares both within-subjects and between-groups. Each participant experienced only one of the two conditions, resulting in a between-subjects design. Each participant completed two rounds in each condition, and their scores were compared not only across conditions but also across rounds, resulting in a within-subjects design in which they competed against themselves. The experimental condition was blind to the leaderboard in the first round, whereas the control condition was aware of the leaderboard throughout both rounds.

Data Analysis: Exploratory behavior was measured by the number of different areas clicked on the grid, and exploitative behavior was measured by the number of times they clicked the same square. We also measured how their behavior changed from round one to round two, based on their first-round score and the distance of that score from the maximum total score. We used a mixed-design ANOVA and Pearson's correlation to analyze our data and calculated effect sizes.

Results: Contributed by: Emma



A



B

Fig. 1: Average scores and number of unique clicks by condition and round. (A) Bar graph of average scores for round 1 of condition 1, $M(SD) = 181.25 (31.04)$, round 1 of condition 2, $M(SD) = 184.84 (40.90)$, round 2 of condition 1, $M(SD) = 227.22 (79)$, and round 2 of condition 2, $M(SD) = 213.87 (70.84)$. (B) Bar graph of average unique clicks of round 1 of condition 1, $M(SD) = 16.17 (2.59)$, round 1 of condition 2, $M(SD) = 14.87 (3.87)$, round 2 of condition 1, $M(SD) = 12.75 (6.24)$, and round 2 of condition 2, $M(SD) = 13.10 (6.22)$.

The average scores for round 1 of conditions 1 and 2 were 181.25 points and 184.84 points, respectively. The average scores for round 2 of conditions 1 and 2 were 227.22 and 213.87 points, respectively. The average number of unique clicks in round 1 for conditions 1 and 2 were 16.17 and 14.87 clicks. The average number of unique clicks in round 2 for conditions 1 and 2 were 12.75 and 13.10 clicks. From this data, we noticed that the average scores in round 2 for condition 1 were slightly higher than round 2 scores in condition 2. Using a mixed-design ANOVA, we tested for main effects and interactions between conditions and rounds.

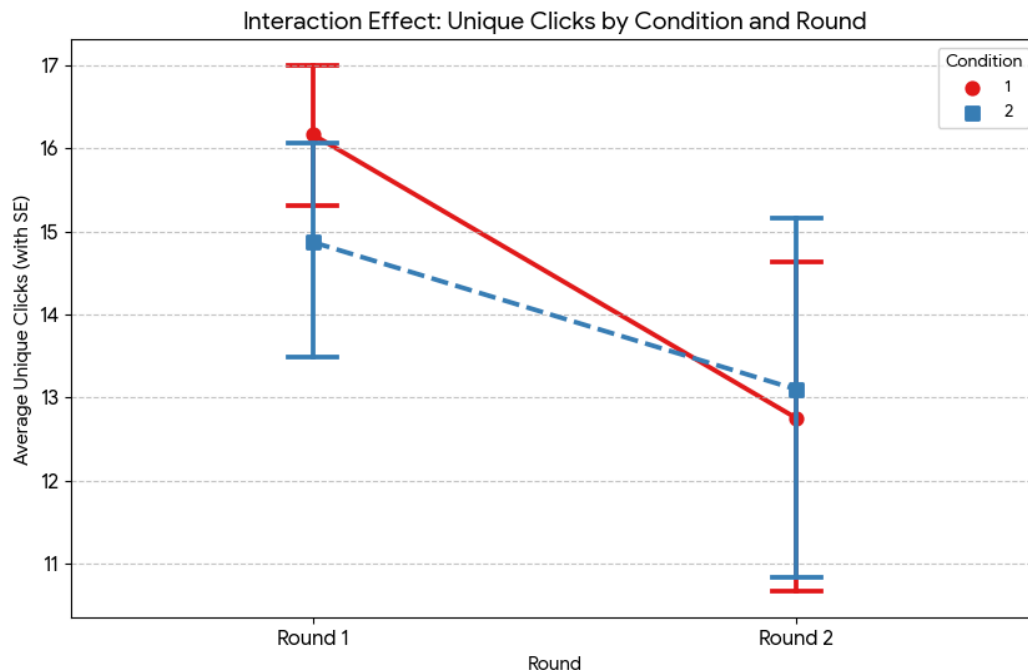


Fig. 2: Mixed-ANOVA results of unique clicks by condition and round. The main effect of Round, $F(1, 65) = 11.93, p < 0.001$, was statistically significant. The main effect of Condition, $F(1, 65) = 0.26, p = 0.61$, was insignificant. The interaction effect (Condition x Round) was also nonsignificant, $F(1, 65) = 1.13, p = 0.29$.

There was a significant main effect between the first round and the second round/control rounds, based on clicking behavior. Specifically, both conditions reduced the number of unique clicks in round 2. This was a large effect, $\eta^2 = 0.155$, which partially confirms our hypothesis that those who are unaware of the leaderboard will exhibit more exploitative behavior in round 2 compared to round 1 because, after viewing the leaderboard, participants in condition 1 reduced the number of unique clicks. Although there was no significant main effect of conditions or interaction effect between conditions and rounds, meaning that both conditions had similar reductions in the number of unique clicks and did not significantly differ in the number of unique clicks in rounds 1 and 2. This is why we can only partially accept/reject our main hypothesis that the leaderboard-cognizant condition would exhibit more exploitative behavior in round 1 compared to the unaware condition. This hypothesis was based on previous research using visual stimuli, which found that more exploitative behavior was used under similar conditions of extrinsic motivation.

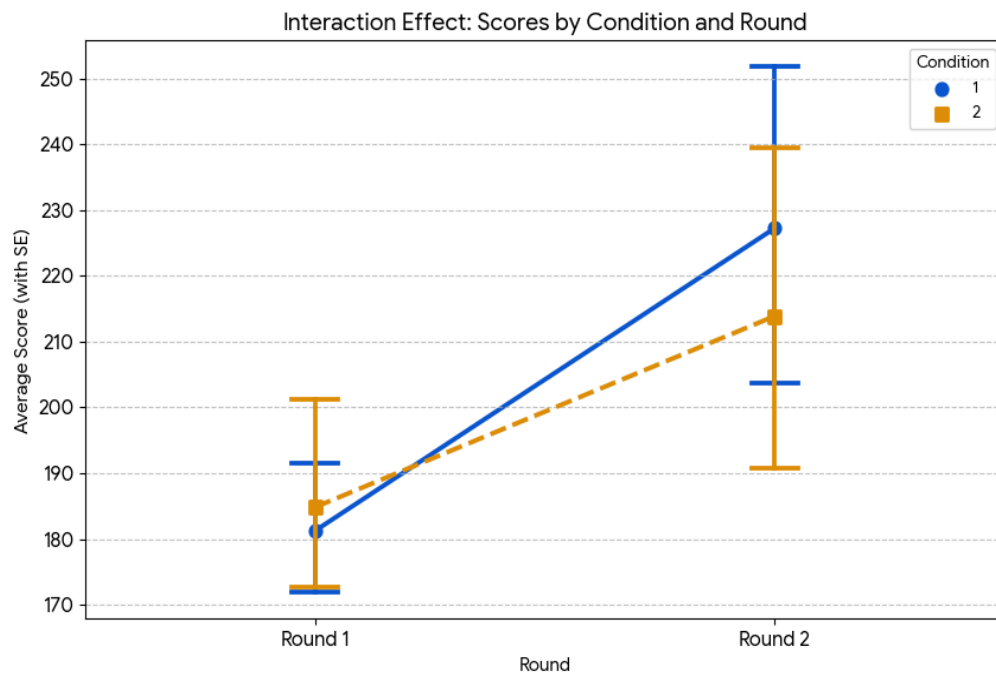


Fig. 3: Mixed-design ANOVA results of scores by condition and round. The main effect of Round, $F(1, 65) = 17.37$, $p < 0.001$, was significant. The main effect of Condition, $F(1,$

65) = 0.19, $p = 0.66$, was nonsignificant. The interaction effect (Condition x Round) was nonsignificant, $F(1, 65) = 0.85$, $p = 0.36$.

There was a significant main effect of rounds one and two on score, with a large effect size, $\eta p^2 = 0.211$. This means that both groups had significantly improved scores from round 1 in round 2. There was no main effect between conditions and no main interaction effect between the conditions and rounds. This means that the conditions did not differ by score. Although condition 1 improved slightly more than condition 2, this difference was not statistically significant given the data's variance. Additionally, we ran a Pearson's correlation to test our secondary hypothesis, where there was an extremely weak correlation ($r = -0.05$), which falsifies our secondary hypothesis that if a higher score was earned in round 1, then more exploitative behavior or fewer unique clicks would be observed in round 2.

Although this study has shown a pronounced learning effect between rounds 1 and 2, the lack of difference between conditions was unexpected due to previous research. Some reasons for this may be that we had a relatively small sample size of only 67 participants and some of our instructions may have been unclear. For example, we did not mention that participants were able to click on the same square multiple times explicitly in the instructions, therefore participants may not have known they were able to do that, which may have skewed our results.

Neuroscience Extensions *Contributed by: Samantha, Emma, and Quincy*

Although there are many different neural processes involved in decision-making¹¹, to measure exploration vs exploitation in the brain, the main region of study is the striatum, which measures reward values and updates action values based on prior experiences^{12,13}. The main signal to study would be dopamine, as it encodes reward prediction errors and uncertainty^{14,15}. The technique would be a functional MRI. The neural prediction generated by our hypothesis is that with the presence of an interactive extrinsic motivator, such as the knowledge of a leaderboard, the striatum will be more active because it encodes the value of scoring higher on the leaderboard. Conversely, the striatum will be less active in the condition in which participants are unaware of the leaderboard because there is no direct social reward.

References Contributed by: Samantha

1. Gupta, A. K., Smith, K. G., & Shalley, C. E. The interplay between exploration and exploitation. *Academy of Management Journal*, **49**, 693-706 (2006).
2. Sutton, R. S., & Barto, A. G. *Reinforcement learning: An introduction*. MIT Press, (1998).
3. Blanco, N. J., & Sloutsky, V. M. Exploration, exploitation, and development: Developmental shifts in decision-making. *Child Development*, **95**, 1287-1298 (2024).
4. Harms, M. B., Xu, Y., Green, C. S., Woodard, K., Wilson, R., & Pollak, S. D. The structure and development of explore-exploit decision making. *Cognitive Psychology*, **150**, (2024).
5. Hester, T., Lopes, M., & Stone, P. Learning exploration strategies in model-based reinforcement learning. *AAMAS '13: Proceedings of the 2013 international conference on Autonomous agents and multi-agent systems*, 1069-1076 (2013).
6. Dubois, M., Bowler, A., Moses-Payne, M. E., Habicht, J., Moran, R., Steinbeis, N., Hauser, T. U. Exploration heuristics decrease during youth. *Cognitive, Affective, & Behavioral Neuroscience*, **22**, 969-983 (2022).
7. Sumner, E., Steyvers, M., & Sarnecka, B. W. It's not the treasure, it's the hunt: Children are more explorative on an explore/exploit task than adults, *Proceedings of the Annual Meeting of the Cognitive Science Society*, 2891-2897 (2019).
8. Li, H. Instructional framing and task ambiguity in motivated learning: A quantum-inspired experimental study on goal superposition and task engagement. *Learning and Motivation*, **93**, (2026).
9. Hacques, G., Komar, J., Dicks, M., & Seifert, L. Exploring to learn and learning to explore. *Psychological Research*, **85**, 1367-1379 (2021).
10. Somerville, L. H., Sasse, S. F., Garrad, M. C., Drysdale, A. T., Abi Akar, N., Insel, C., & Wilson, R.C. Charting the expansion of strategic exploratory behavior during adolescence. *Journal of Experimental Psychology: General*, **146**, 155-164 (2017).

11. Hartley, C. A., & Somerville, L. H. The neuroscience of adolescent decision-making. *Current Opinion in Behavioral Sciences*, **5**, 108-115 (2015).
12. Samejima, K., Ueda, Y., Doya, K., & Kimura, M. Representation of action-specific reward values in the striatum. *Science*, **310**, 1337-1340 (2005).
13. Kim, H., Sul, J. H., Huh, N., Lee, D., & Jung, M. W. Role of striatum in updating values of chosen actions. *The Journal of Neuroscience*, **29**, 14701-14712 (2009).
14. Mohebi, A., Pettibone, J. R., Hamid, A. A., Wong, T. J.M., Vinson, L. T., Patriarchi, T., Tian, L., Kennedy, R. T., & Berke, J. D. Dissociable dopamine dynamics for learning and motivation. *Nature*, **570**, 65-70 (2019).
15. Schultz, W. Dopamine signals for reward value and risk: basic and recent data. *Behavioral and Brain Functions*, **6**, (2010).

Contributor Block

Abstract: Samantha (Primary), Quincy, Emma

Introduction: Samantha

Methods: Quincy (Primary), Emma, Samantha

Expected Results/Data Analysis: Emma

Neuroscience Extensions: All members equally

References: Samantha

Programming & Code Design: Quincy and Emma

Presentation: All members equally

All group members reviewed and approved each section.